

UKHTA, Russian Federation

WMO: 236060

Lat: 63.5753N

Lon: 53.8058E

Elev: 137

StdP: 99.69

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.2	-32.8	-39.2	0.1	-36.2	-35.6	0.1	-32.7	7.9	-8.2	7.0	-10.9	1.2	20	0.705

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	27.9	18.8	25.8	17.9	23.6	16.7	20.2	25.7	18.9	23.8	17.7	22.3	2.7	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.2	13.3	22.1	17.1	12.4	21.1	15.8	11.4	20.3	58.6	25.8	54.3	23.8	50.5	22.4	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 7.9	6.7	6.2	DB	-37.4	30.7	3.8	2.3	-40.1	32.3	-42.4	33.7	-44.5	34.9	-47.2	36.6	(4)
(5)			WB	-37.4	21.8	3.7	1.4	-40.1	22.8	-42.3	23.6	-44.4	24.4	-47.1	25.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.2	-16.2	-14.9	-7.3	-0.1	6.5	12.6	16.2	12.7	7.5	0.6	-8.0	-12.7	(6)	
	DBStd	12.69	8.90	8.60	6.00	5.56	5.80	5.08	4.66	4.22	3.83	4.50	7.54	9.01	(7)	
	HDD10.0	4180	814	697	536	306	143	29	6	18	94	293	539	704	(8)	
	HDD18.3	6812	1072	931	794	554	369	182	97	183	326	551	789	962	(9)	
	CDD10.0	463	0	0	0	2	34	108	199	102	18	1	0	0	(10)	
	CDD18.3	54	0	0	0	0	2	12	32	8	0	0	0	0	(11)	
	CDH23.3	497	0	0	0	0	32	123	286	56	1	0	0	0	(12)	
	CDH26.7	112	0	0	0	0	5	28	67	12	0	0	0	0	(13)	
(14) Wind	WSAvg	3.1	2.9	2.9	3.2	3.3	3.5	3.2	2.7	2.7	3.0	3.2	3.2	3.2	(14)	
(15) Precipitation	PrecAvg	581	36	29	32	34	46	72	61	69	60	59	45	38	(15)	
	PrecMax	723	55	51	56	91	86	179	178	136	133	105	76	73	(16)	
	PrecMin	476	19	9	5	2	8	24	17	12	22	30	11	24	(17)	
	PrecStd	61	10	12	14	20	20	30	34	27	25	23	14	11	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.2	1.6	7.2	17.7	27.1	29.9	31.1	28.6	20.8	13.8	6.8	2.2	(19)	
		MCWB	-0.2	0.5	2.4	9.3	16.1	19.2	20.7	20.1	15.3	9.8	6.4	1.5	(20)	
	2%	DB	-1.1	0.1	3.6	12.6	23.2	26.9	28.5	24.5	17.2	10.2	3.0	0.7	(21)	
		MCWB	-1.4	-0.7	0.4	6.7	14.0	18.0	19.4	17.7	12.8	8.6	2.6	0.2	(22)	
	5%	DB	-2.9	-1.5	1.9	9.5	20.0	24.1	26.4	22.1	15.0	8.4	1.7	-0.2	(23)	
		MCWB	-3.3	-2.2	-0.1	4.6	12.1	16.5	18.6	16.4	12.1	7.2	1.3	-0.6	(24)	
	10%	DB	-5.0	-3.2	0.6	7.1	16.8	21.4	24.5	19.7	13.1	6.7	0.4	-1.7	(25)	
		MCWB	-5.5	-3.9	-0.8	3.3	10.2	14.9	17.8	15.4	11.0	5.8	0.1	-2.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.1	0.8	2.9	10.1	16.8	20.7	22.0	20.8	15.9	11.0	6.0	1.5	(27)	
		MCDB	0.1	1.3	6.2	16.6	23.4	26.3	27.4	26.2	20.0	13.0	6.6	1.8	(28)	
	2%	WB	-1.6	-0.7	1.2	6.9	14.7	19.0	20.8	18.9	13.8	8.7	2.5	0.1	(29)	
		MCDB	-1.3	-0.1	2.7	11.7	22.1	24.5	26.5	23.3	16.2	10.0	2.7	0.4	(30)	
	5%	WB	-3.4	-2.3	0.2	5.2	12.6	17.5	19.6	17.5	12.5	7.2	1.2	-0.7	(31)	
		MCDB	-3.0	-1.6	1.5	8.4	19.0	22.8	24.6	20.8	14.6	8.2	1.4	-0.3	(32)	
	10%	WB	-5.5	-3.9	-0.7	3.8	10.9	15.8	18.4	16.0	11.2	5.7	0.1	-2.2	(33)	
		MCDB	-5.0	-3.2	0.7	6.4	16.0	20.4	23.3	18.8	13.0	6.5	0.3	-1.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.5	7.1	8.6	8.6	9.7	9.6	9.9	8.3	6.5	4.3	5.0	5.8	(35)	
		MCDBR	6.9	5.1	8.1	11.6	14.5	13.0	12.8	11.9	10.1	6.3	3.9	4.9	(36)	
		MCWBR	6.9	4.8	6.1	7.4	7.8	6.4	5.8	6.0	6.2	5.1	4.0	4.9	(37)	
	5% WB	MCDBR	6.7	5.1	6.5	9.8	13.5	11.6	11.6	10.5	8.7	5.8	3.6	4.7	(38)	
		MCWBR	6.7	5.1	5.4	6.8	7.9	6.5	6.0	6.0	6.0	5.0	3.8	4.8	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.236	0.285	0.320	0.352	0.354	0.362	0.410	0.377	0.337	0.307	0.235	0.226	(40)	
	taud		1.929	2.038	2.055	2.105	2.383	2.440	2.306	2.381	2.470	2.318	1.996	1.915	(41)	
	Ebn at Noon		329	610	748	814	858	858	795	793	758	608	330	136	(42)	
	Edh at Noon		32	69	103	123	103	100	111	94	71	53	28	15	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.20	0.90	2.29	4.02	5.01	5.18	5.20	3.50	1.88	0.72	0.21	0.08	(44)	
	RadStd		0.02	0.13	0.28	0.43	0.42	0.45	0.47	0.40	0.25	0.09	0.05	0.01	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d) +0.98	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (2 neighbors)	+0.98	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page